DAY 2 – NOTEBOOK EXECUTION GUIDE

Hospital Patient Journey and Operations Data Engineering Project | Silver Layer

Purpose

This guide explains which code belongs in which Databricks notebook and the order in which the notebooks should be run. It is based on the Day 2 Silver Layer implementation guide. The four Day 2 notebook names are practical suggestions, not mandatory names specified by the Day 1 reference.

1. Reuse the existing configuration notebook

Notebook: 00_config | **Status:** Existing Day 1 notebook – reuse it; do not create another one.

This notebook is the single source of truth for catalog names, schema names, paths and table definitions. Confirm that its values match your assigned environment.

At the beginning of every new Day 2 notebook, run:

```
%run ./00_config
```

2. Day 2 notebooks and their code

Notebook 1 – 10_silver_events

Purpose: Transform the primary encounter Bronze table into clean, typed Silver data.

Code sections – run in this order:

1. Import PySpark functions, Window and required data types; read the primary Bronze table.
2. Preserve raw copies of fields and required source-lineage columns.
3. Standardize strings and cast timestamps, decimal and integer fields.
4. Replace missing visit_type values with UNKNOWN and create a missing-value flag.
5. Create validation flags for invalid timestamps, bill amounts and wait times.
6. Deduplicate by encounter_id using latest _source_file_name, then _source_row_number.
7. Calculate the extreme bill threshold using Q3 + 3 × IQR and flag extreme values.
8. Separate valid and rejected records; attach explicit rejection reasons.
9. Write valid encounter records and rejected records to the approved Silver tables.

Expected outputs: silver_encounter_events; silver_rejected_records (proposed supporting table)

Refer to: Day 2 PDF pages 3–5

Notebook 2 – 11_silver_cdc

Purpose: Process status updates and derive the latest status for each encounter.

Code sections – run in this order:

1. Read the status-updates Bronze table and inspect its schema.

2. Preserve raw values and cast change_timestamp and source_sequence; standardize status strings.

3. Validate timestamps, sequence values and encounter IDs against primary Silver encounters.

4. Write valid status records to silver_status_history.

5. Use a Window partitioned by encounter_id and ordered by change_timestamp and source_sequence to select the latest row.

6. Write the latest-status representation to silver_encounter_current.

7. Prepare and execute synthetic TC04 tests separately from production Bronze data.

Expected outputs: silver_status_history; silver_encounter_current

Refer to: Day 2 PDF pages 5–6

Notebook 3 – 12_silver_reference_events

Purpose: Prepare department, patient and secondary-event datasets for Silver-layer use.

Code sections – run in this order:

1. Read department and patient Bronze tables; inspect their schemas and sample rows.

2. Apply agreed trimming, uppercasing and type rules while retaining raw copies.

3. Read the secondary-events Bronze table.

4. Standardize encounter_id and event_type; preserve raw values.

5. Cast event_timestamp and event_value where those fields exist.

6. Apply required-field validation and write valid secondary-event data using an approved table name.

Expected outputs: Silver department and patient outputs (names to be confirmed); Approved secondary-event Silver output

Refer to: Day 2 PDF page 7

Notebook 4 – test_day2_silver

Purpose: Run integration checks and collect evidence for all Day 2 outputs.

Code sections – run in this order:

1. Verify the five Bronze table counts: 11,540 primary events; 520 status updates; 310 secondary events; 110 departments; 150 patients.

2. Check Silver row counts and distinct encounter IDs.

3. Confirm 40 surplus duplicate rows were handled and 11,500 encounter IDs remain before other rejection effects.

4. Confirm 12 missing visit_type values were replaced with UNKNOWN and flagged.

5. Confirm 35 extreme bill values are flagged and retained.

6. Check rejection reasons and required lineage columns.

7. Validate status-history counts and current-status output.

8. Check secondary-event counts: ALERT 98, REVIEW 118 and CHECK 94.

9. Run synthetic TC04 cases, capture screenshots and record unresolved decisions.

Expected outputs: Validation outputs and screenshots; no new business table is necessarily required

Refer to: Day 2 PDF pages 7–8

3. Correct execution sequence

Order	Notebook	Action
1	00_config	Reuse existing Day 1 configuration
2	10_silver_events	Build primary encounter Silver outputs
3	11_silver_cdc	Build status history and current status
4	12_silver_reference_events	Prepare reference and secondary-event data
5	test_day2_silver	Run validation and capture evidence

4. Important instructions before running

• Start each new Day 2 notebook with **%run ./00_config**.

• Replace placeholders such as <catalog>, <bronze_schema> and <silver_schema> with exact values from 00_config. Use the exact Bronze table names and column names from your workspace.

• Do not modify, clean or overwrite Bronze tables. Day 2 transformations must read from Bronze and write to Silver.

• If a Bronze count differs from the reference count, investigate Day 1 ingestion before continuing.

• Do not invent status mappings or priority mappings. The Day 2 guide marks these decisions as open.

• Do not implement masking or expose restricted identifiers until the instructor or workspace administrator approves the access-control approach.

• Overwrite examples are for a controlled first development build only. Reruns and production need a deliberate refresh or MERGE strategy.

• Confirm proposed supporting table names with the instructor before creating them.

5. Evidence to capture

- Five Bronze tables and verified row counts.
- Primary Silver schema, typed fields, raw copies and lineage columns.
- Duplicate reconciliation and deduplicated output count.
- UNKNOWN visit_type count and missing-value flag.
- Extreme bill flag count (expected 35).
- Rejected-record schema and rejection reasons.
- Status history and latest-status output.
- Secondary-event counts by event_type.
- Approved role-based masking/access test results, or a recorded blocker.
- Synthetic TC04 test results and notebook screenshots.

Source note: This execution map summarizes the uploaded Day 2 Silver Layer guide. The guide states that the proposed notebook names and some supporting output names should be confirmed against the instructor's rubric or project worksheet.